Program 1: Design a Professional Navigation Bar

CSS concepts: Background, padding, hover effect, inline navigation

Client requirement: A client wants a professional navigation menu for a college, business or personal website. The menu should contain Home, About, Services and Contact. When the user moves the mouse over an item, its appearance should change.

Student task: Apply CSS to the existing website's navigation bar.

Explanation:

- `background-color` sets the menu's background.
- `padding` adds space around the menu items.
- `text-decoration: none` removes the underline.
- `:hover` applies a different style when the mouse moves over a link.

Expected output: A dark navigation bar with white links that change appearance on hover.

Program 2: Create a Product or Service Card

CSS concepts: Borders, shadows, margins, padding and buttons



Client requirement: An online shopping client wants products displayed in attractive cards with a product name, price, description and Buy Now button.

Student task: Add a product card to an existing shopping or service website.

Explanation:

- `border-radius` makes the card's corners rounded.
- `box-shadow` adds a shadow around the card.
- `margin: 30px auto` centers the card horizontally.
- `.price` styles only the price, while `.product button` styles the button.

Expected output: A centered product card with a price and interactive-looking button. The button is for styling only; purchasing functionality would require additional code.

Program 3: Design a Responsive Website Layout

Client requirement: A company wants its website to display three service sections side by side on a computer but stack them vertically on a mobile phone.

Student task: Make the existing website responsive without creating a separate mobile website.

Explanation:

- `display: flex` arranges the sections in a row.
- `flex: 1` allows the sections to share the available space.
- `@media (max-width: 600px)` applies special styling when the screen width is 600 pixels or less.
- `flex-direction: column` stacks the sections vertically.

Expected output: Three columns on a wider screen and vertically stacked sections on a narrow screen.

Program 4: Design a Student Registration Form

Client requirement: A college wants a student registration form that looks professional and is easy to use.

Student task: Apply CSS to the form on an existing admission or registration webpage.

Explanation:

- `max-width` prevents the form from becoming excessively wide.
- `box-sizing: border-box` keeps the input width within the form.
- `:focus` highlights the field currently selected by the user.
- `:hover` changes the button's background when the mouse moves over it.

Expected output: A centered registration form with neatly aligned input fields.

Program 5: Create a Website Header and Hero Section

CSS concepts: Background image, text alignment, typography and buttons



Client requirement: A college wants an attractive homepage featuring a welcome message, a short description and an admission button.

Student task: Improve the homepage of an existing college, travel, hotel or business website.

Explanation:

- `min-height` sets the minimum height of the hero section.
- `linear-gradient()` creates a gradual transition between background colors.
- `text-align: center` centers the content.
- `display: inline-block` allows the link to be styled like a button.

Expected output: A large, colored homepage banner with a heading, description and Apply Now link.

Extension task: Ask students to replace the gradient with a relevant background image using `background-image`.

Program 6: Design a Pricing Table for a Business Website

CSS concepts: Flexbox, borders, spacing, typography and hover effects

Client requirement: A web development company wants to show three service packages: Basic, Standard and Premium.

Student task: Create pricing cards for an existing business website.

Explanation:

- `display: flex` places the pricing cards side by side when space allows.
- `flex-wrap: wrap` lets cards move to the next row when space is limited.
- `transition` makes the hover animation smoother.
- `transform: translateY(-5px)` moves the selected card upward by five pixels.

Expected output: Three pricing cards that respond visually when the mouse moves over them.

Program 7: Create a Consistent Website Theme Using CSS Classes

CSS concepts: External CSS, reusable classes, selectors and consistent styling

Client requirement: A company already has five web pages. The client wants all pages to use the same font, colors, headings, buttons and footer. The student must avoid writing the same CSS separately on every page.

Student task: Create a common CSS file and link it to the existing HTML pages.

Explanation:

- `<link rel="stylesheet" href="style.css">` connects the external CSS file to the webpage.
- `.btn` is a reusable class that can style multiple buttons or links.
- The `header` and `footer` selectors provide consistent styling across the website.
- The same `style.css` file can be linked to About, Services, Contact and other HTML pages.

Expected output: A consistent website theme that can be maintained from a single CSS file.

Extension task: Ask students to change the main theme color in `style.css` and observe how the design changes across all linked pages.